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Data Science Technologies

Assignment #3

ID	Gender	Study Hours	Attendance	Result
1	Male	2	Low	Fail
2	Female	5	High	Pass
3	Male	6	High	Pass
4	Female	1	Low	Fail
5	Male	4	High	Pass

Classify This using Naive Bayes:

Gender	Study Hours	Attendance
Female	3	High

Gender	F	P
Male	1	2
Female	1	1

Study Hours	F	P
2	1	0
5	0	1
6	0	1
1	1	0
4	0	1

Attendance	F	P
low	2	0
High	0	3

Fail	Pass
2	3

1/2	2/3	1/2	0/3	2/2	0/3
1/2	1/3	0/2	1/3	0/2	3/3
		0/2	1/3		
		1/2	0/3		
		0/2	1/3		

Fail = $1/2 \times 0/2 \times 0/2 \times 2/5 = 0$

Pass = $1/3 \times 0/3 \times 3/3 \times 3/5 = 0$

$P(E) = 0 + 0 =$

Therefore,

$P(Pass/E) = 0/0$

$P(Fail/E) = 0/0$

This is undefined.

→ If final class is undefined use the Laplace smoothing:

	Pass	Fail
Prior	3/5	2/5
Gender = Female	$(1+1)/(3+2) = 2/5$	$(1+1)/(2+2) = 2/4$
StudyHours = 3	$(0+1)/(3+6) = 1/9$	$(0+1)/(2+6) = 1/8$
Attendance = High	$(3+1)/(3+2) = 4/5$	$(0+1)/(2+2) = 1/4$

Pass Score = $3/5 \times 2/5 \times 1/9 \times 4/5 = 0.02133$

Fail Score = $2/5 \times 2/4 \times 1/8 \times 1/4 = 0.00625$

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As $0.02133 > 0.00625$

So The class after Laplace smoothing is:
"Pass"